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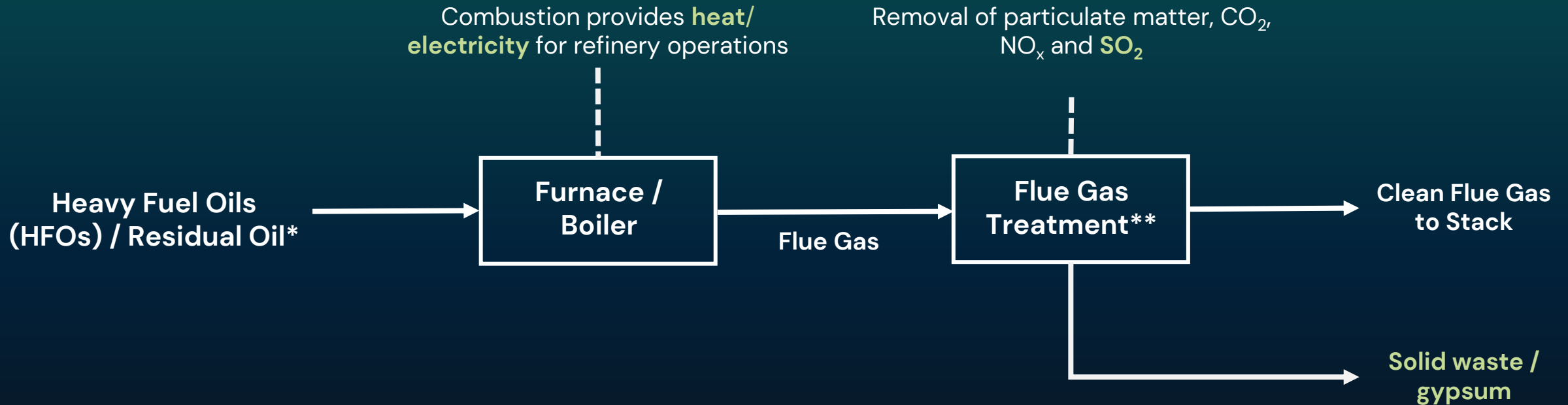


Co-Production of Green Hydrogen and Potassium Sulphate from Flue Gas SO₂ Emissions

Lord of the Rigs

Aayush Gogate, Arko Das, Emil Medzhidov, Georgios Papageorgiou, Yan Tat Loo

Conventional Flue Gas Desulphurisation



* Up to 3.5 %wt oil sulphur content (IMO, 2021)

** Continuous reagent feed (CaCO₃)



FGD is a **\$22B** industry built entirely on ***burying a resource***

– Future Market Insights, 2025

Problem Statement



“Wet scrubbers represent more than **80% of FGD** units, often producing **low-value byproducts** for disposal or sale”

– Sorrels, J.L., 2021

“How can this waste stream be exploited to create a more **high-value, renewable product?**”

– The Lord of the Rigs, 2026

Proposed Solution



Integrated **electrochemical engine** converting SO_2 into **dual-stream** revenue of **green H_2** and **K_2SO_4 fertiliser**

Proposed Solution



Energy Efficient

Generates green H_2 at **85% lower voltage** than **water splitting** (Ün, 2007)



Market Driven

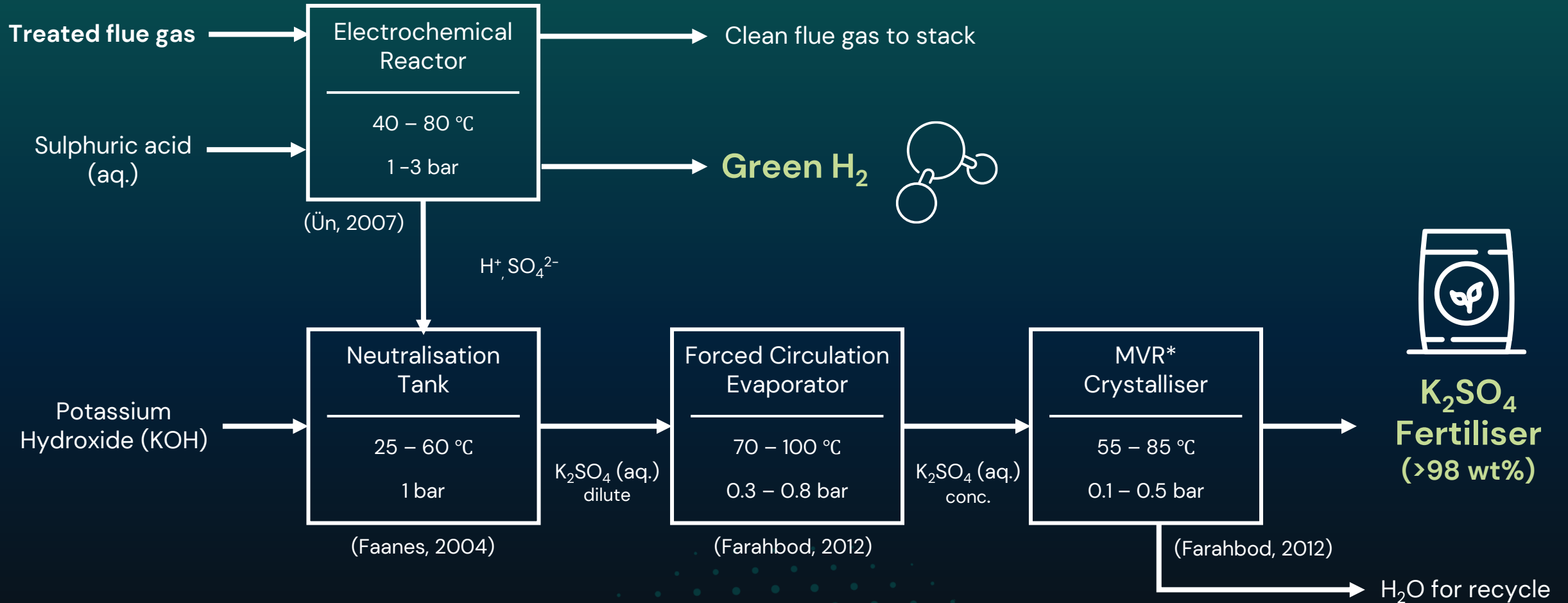
Chloride free K_2SO_4 for **high-value agriculture**



Zero Waste

Closed-loop system with **zero water discharge (ZLD)**

Proposed Solution



*Mechanical Vapor Recompression

Proposed Solution

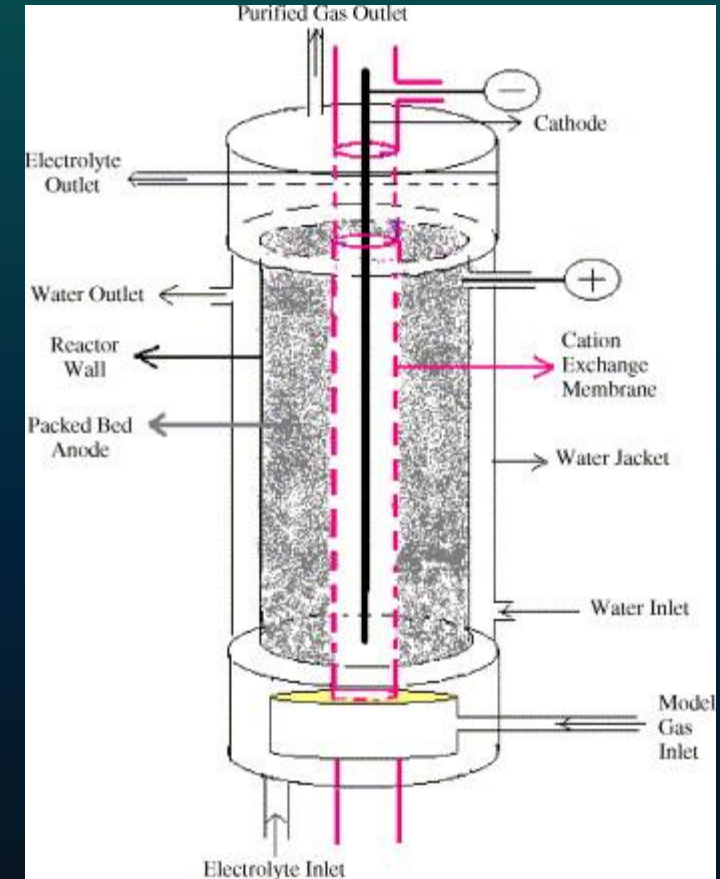
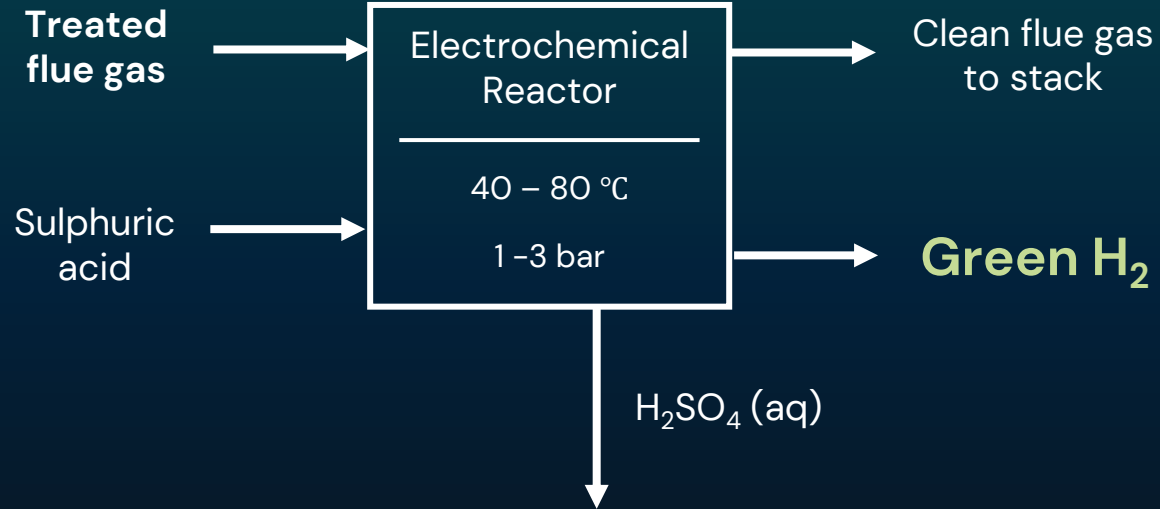
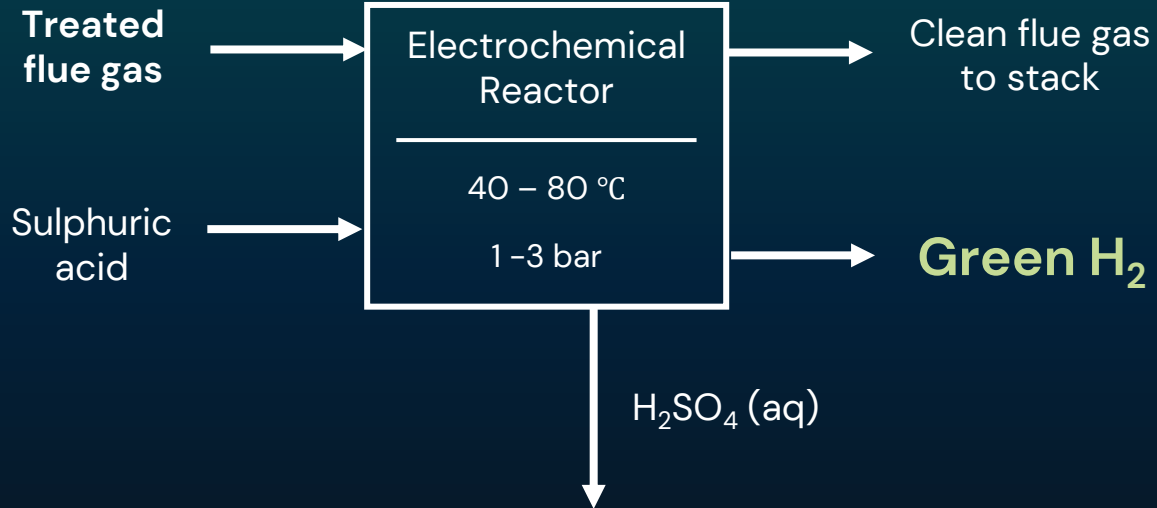
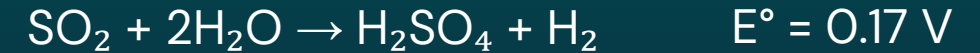


Fig. 1 Electrochemical reactor (Ün, 2007)

Proposed Solution

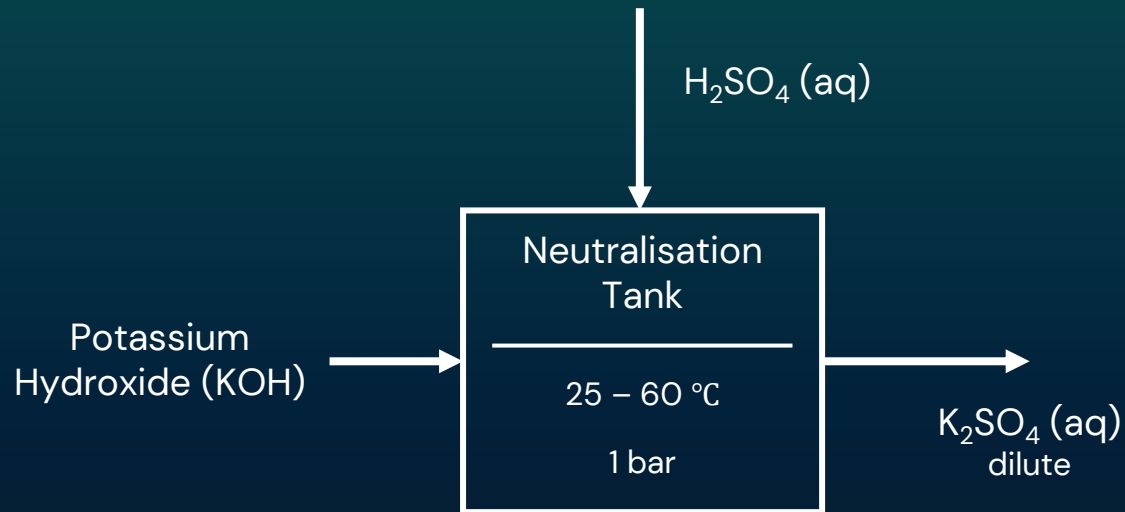


Net Reaction



- 200,000 Nm³/hr @ 1,000 ppm SO₂ flue gas used
- SO₂ conversion **>90%** (Ün, 2007)
- **Platinum mesh** catalyst
- **30 wt% H₂SO₄** – balance between SO₂ solubility and electrolyte conductivity

Proposed Solution



Reaction



Potential side reaction



- **Reaction** goes to **completion** due to strength of the acid/base
- **KOH** fed as solution, slight **molar excess** to ensure completion (2.05:1)
- Typical residence of **15 minutes** (Green et. al, 2019)
- High-Grade **Stainless Steel 316L**

Integrated Safety & Risk Management



Hazards



H₂ Explosion

4 – 75% LFL



Thermal Runaway

Neutralisation (–114 kJ/mol)



Accidental SO₂ Release

Respiratory & acid rain
(DEFRA, 2024)

Mitigations

01

Inherently Safer Design (ISD)

Limited to **sub-boiling temperatures** and **low-pressure differentials**

(Sustainability Directory, 2025)

02

Active Automation

Real-time monitoring with **Automated Fail-Closed Interlocks**

(HSE UK, n.d.)

03

Mechanical Failsafes

Pressure relief valves (PRV) and rupture disks

(Barton et. al, 2007)

Holistic Sustainability & Circularity



Waste-to-Value

SO₂ emissions to premium SOP **fertilisers**



Zero Liquid Discharge

100% water recovery



Energy Integration

Heat integration potential between exotherm and feed



Carbon Offset Potential

Zero Scope 1 emission (Sand et al, 2023) with lifecycle CO₂ offset

Economics Feasibility



Expenditure

£11 Million

CAPEX

£7 Million

OPEX

Revenues

£ 10 Million

K₂SO₄ fertiliser

£0.7 Million

Green H₂

£ 18 Million

NPV

16.9%

ROI

-£145/t SO₂

Net Levelised Cost

1. Values are in order of magnitude estimation ($\pm 50\%$)
2. Lang factor 4.74 on major equipment (Sinnott, 2019)
3. K₂SO₄ fertilizer £800/t – assumed green premium (IMARC, 2019)
4. Green H₂ £5,000/t – market price (IMARC, 2019)
5. Negative net levelized cost means generates net income
6. NPV assumes 8%, 20 years

*Conventional FGD: ~ £700/t SO₂
(Kaminski, 2003)

Implementation Roadmap



01

Proof of Concept

0 – 6 months

- Validate **electrochemical absorption** at lab scale
- Confirm **H₂ purity** & **SO₄²⁻ yield** from flue gas feed
- Characterise **fertiliser grade** & purity

02

Prototype

7 – 18 months

- Build **bench-scale unit**
- Continuous feed from **simulated** refinery flue gas
- Optimise **KOH stoichiometry** & **cost efficiency**
- **Safety**, materials & regulatory assessment

03

Industrial Field Trial

19 – 36 months

- Partner with refinery for **on-site pilot**
- Integration with **live flue gas**
- **Techno-economics** and **life cycle** assessment
- Secure H₂ off-take & K₂SO₄ distribution agreements



THANK YOU

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Appendix – CAPEX

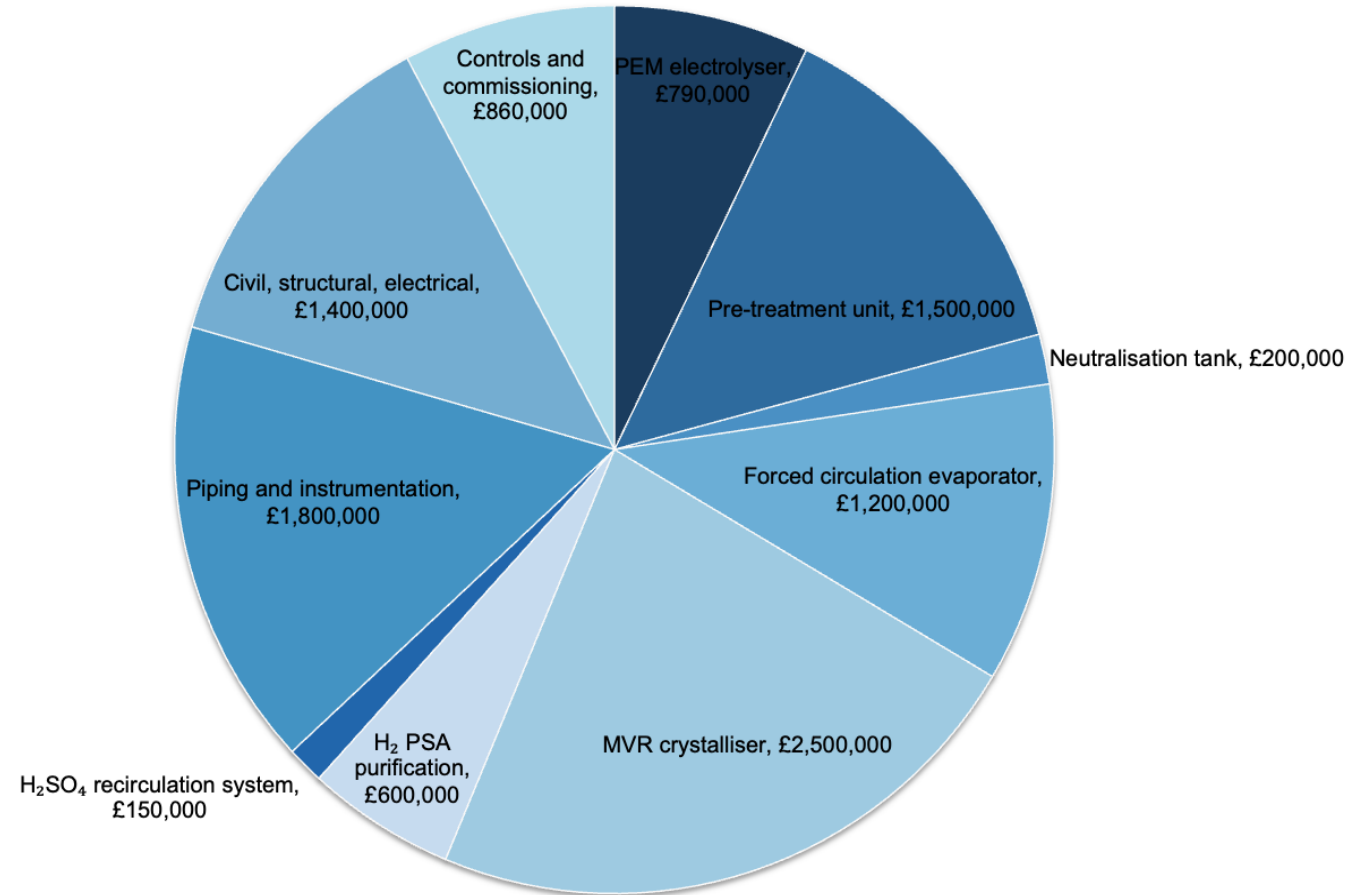


Fig. 2 Estimated CAPEX Breakdown

Appendix – OPEX



Table 1: Order-of-Magnitude OPEX breakdown

Item	Quantity	Unit cost	Annual cost	% OPEX
KOH reagent	7,900 t/year	£800/t	£6,320,000	84%
Electricity	5,755 MWh/year	£60/MWh	£345,000	5%
H ₂ SO ₄ make-up	~500 t/year	£100/t	£50,000	1%
Maintenance	3% of CAPEX	—	£330,000	4%
Labour	4 operators	£100,000/head	£400,000	5%
Water / blowdown treatment	Allowance	—	£80,000	1%
Total OPEX			£7,525,000	100%

Appendix – Profitability



Table 2: Preliminary Profitability Calculations

Metric	Calculation	Value
EBITDA	$£10,505,000 - £7,525,000$	$£2,980,000/\text{year}$
CRF (8%, 20 yr)	$0.08 \times (1.08)^{20} / ((1.08)^{20} - 1)$	0.102
Annual capital charge	$£11,000,000 \times 0.102$	$£1,122,000/\text{year}$
Net annual profit	$£2,980,000 - £1,122,000$	$£1,858,000/\text{year}$
Simple payback	$£11,000,000 / £2,980,000$	5.9 years
NPV (8%, 20 yr)	$(£2,980,000 \times 9.818) - £11,000,000$	$£18,260,000$
ROI	$£1,858,000 / £11,000,000$	$16.9\%/\text{year}$

Appendix – Net Levelised Cost



Table 3: Net-Levelised calculations for comparison

Parameter	Calculation	Value
Total annual cost (gross)	$£1,122,000 + £7,525,000$	$£8,647,000/\text{year}$
Gross levelised cost	$£8,647,000 / 4,500 \text{ t SO}_2$	$£1,922/\text{t SO}_2$
Revenue credit per t SO ₂	$£10,505,000 / 4,500$	$£2,334/\text{t SO}_2$
Net levelised cost	$£1,922 - £2,334$	$-£415/\text{t SO}_2$